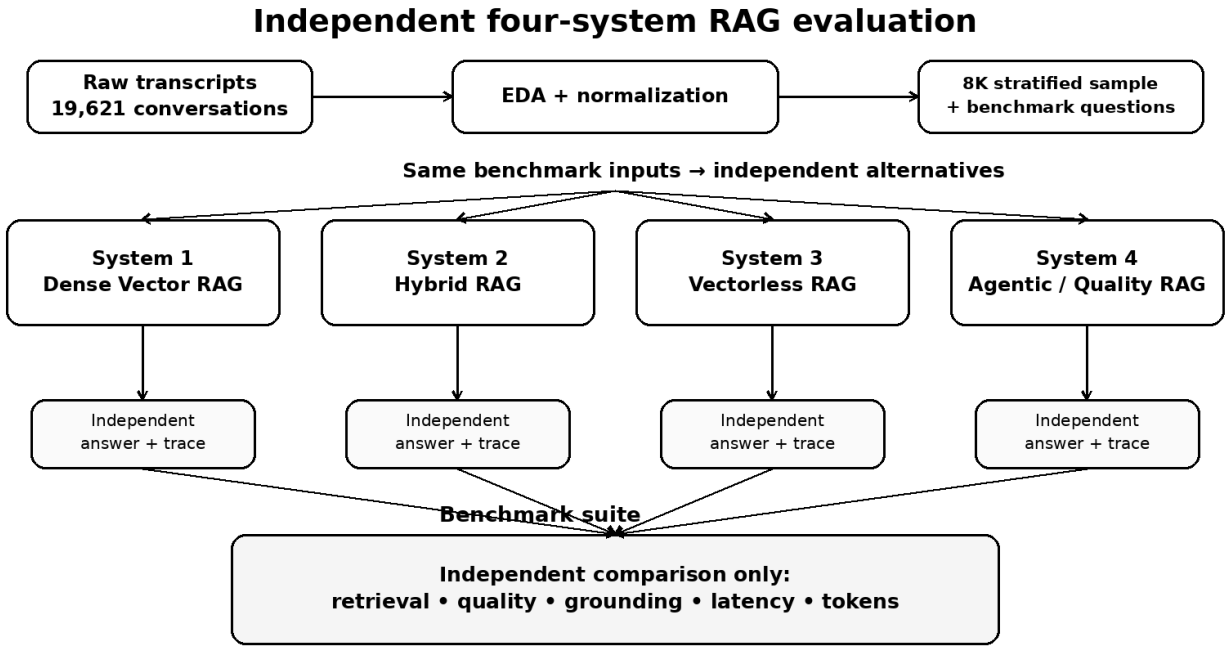
A Multi-Architecture RAG Framework for Causal Rationale Extraction from Conversational Business Data

Anonymous Technical Report

Abstract— This report presents an end-to-end retrieval-augmented generation (RAG) framework for extracting and synthesizing causal rationales from noisy business conversations. The project was developed around a 19,621-conversation, two-party conversational dataset and progressed through four retrieval architectures: dense vector RAG, hybrid retrieval, vectorless lexical RAG, and a multi-stage agentic/quality pipeline. The systems share a common evidence model, grounding checks, structured outputs, and benchmark instrumentation so that retrieval behavior can be isolated from generation behavior. Extensive exploratory analysis was used to characterize conversation length, intent diversity, speaker-label noise, and metadata structure. The latest 13-query benchmark shows a clear trade-off: the hybrid system achieves strong evidence precision and recall but is dominated by LLM generation latency; the vectorless system reduces latency but suffers substantially higher cross-conversation contamination; and the agentic system achieves strong conversation selection and very low measured control-pipeline latency, but currently over-compresses evidence and uses a deterministic fallback rather than a measured LLM generation stage. The report documents the architecture, experiments, failure modes, fixes, and several improvements that remained unfinished because of time constraints.

Index Terms— retrieval-augmented generation, causal rationale extraction, conversational AI, hybrid retrieval, vectorless RAG, reranking, evidence grounding, Laya, local inference.



Project scope. The underlying hackathon dataset is proprietary/copyrighted and is not redistributed with this report. The report uses aggregate statistics and benchmark artifacts only. The system is designed for local-first operation, with Ollama-based Qwen3.5 MLX generation used in the measured System 2 and System 3 runs; System 4's latest benchmark used a deterministic fallback unless an Ollama model is configured.

I. INTRODUCTION

Business conversations contain causal information in a form that is substantially more difficult to retrieve than a conventional fact. The cause may appear in one turn, the customer's motivation in another, an agent intervention several turns later, and the final outcome near the end of the interaction. The project therefore treats RAG as an evidence-construction problem rather than simply a top-k similarity search. A central design requirement is that evidence must remain attached to its originating conversation. This requirement emerged directly from early experiments in which globally retrieving individually similar chunks caused unrelated customers and business events to be synthesized into a single answer.

The work consequently evolved into four controlled systems. System 1 establishes a dense semantic baseline. System 2 combines dense and lexical retrieval. System 3 deliberately removes embeddings and vector databases to test a vectorless alternative. System 4 adds broad candidate retrieval, reranking, causal evidence selection, Laya-controlled gates, compression, and grounding validation. A separate benchmark suite records retrieval quality, contamination, latency distributions, token-related measurements, and failure traces.

II. DATASET AND EXPLORATORY ANALYSIS

The source contains 19,621 English conversations between an agent and a customer, with call-level metadata including transcript identity, interaction time, domain, intent, and reason for call. The dataset intentionally contains real conversational noise: incomplete punctuation, fillers, hesitations, non-grammatical phrases, and inconsistent speaker labels. The official problem statement also notes that metadata other than transcript identity can be null. In the materialized dataset used for this report, the four inspected metadata fields were populated for all 19,621 records.

TABLE I. DATASET PROFILE

Property	Value
Conversations	19,621
Total turns	686,764
Mean turns / conversation	35.00
Median turns / conversation	35
Range of turns	1–79
Unique intents	35
Domains	1 (Insurance)
Interaction dates	9 (2025-11-04 to 2025-11-12)
Raw speaker labels	661
Agent + Customer turn share	94.19%
Mean words / turn	15.73

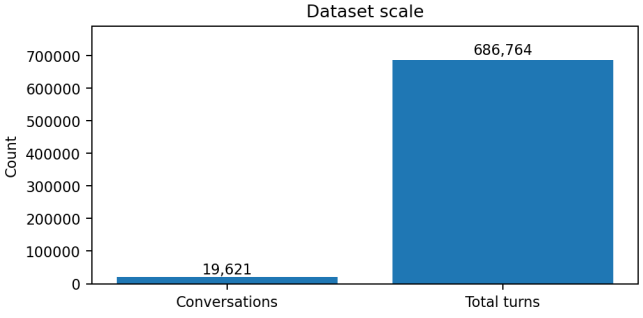


Fig. 2. Dataset scale. The corpus contains 686,764 individual turns across 19,621 conversations.

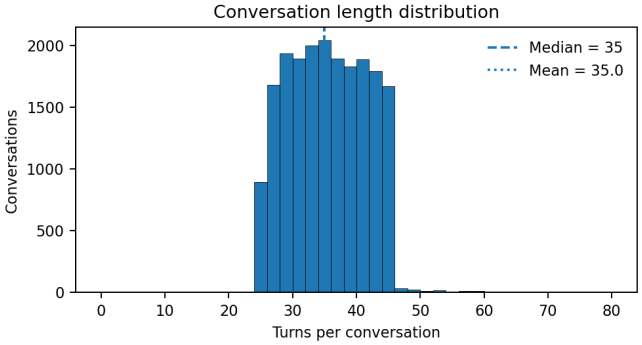


Fig. 3. Conversation length is centered around 35 turns, with a range from 1 to 79.

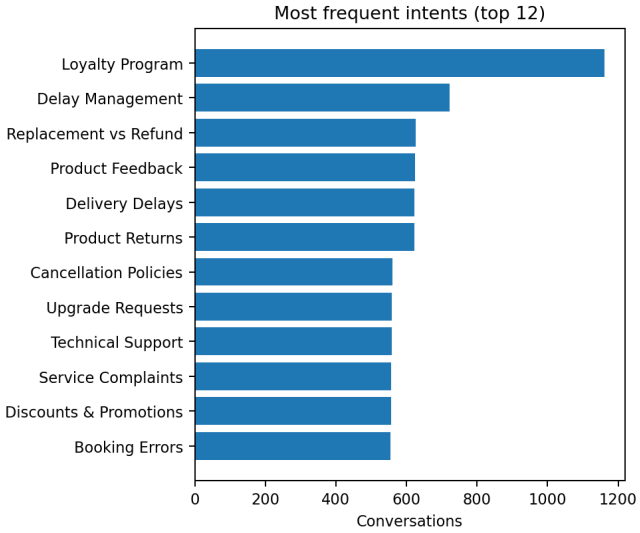


Fig. 4. The dataset spans 35 intents; the largest categories include loyalty, delay, returns/refunds, cancellation, upgrades, and technical support.

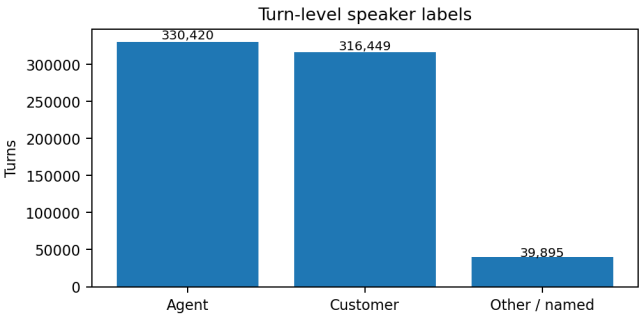


Fig. 5. Most turns use canonical Agent/Customer labels, but the remaining named/special labels motivate defensive speaker normalization.

A. EDA Findings and Modeling Implications

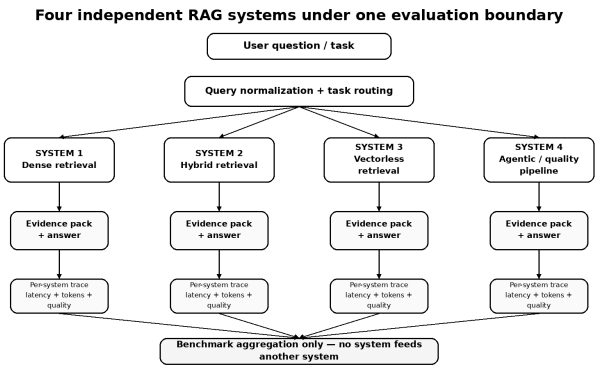
• **Conversation locality is essential.** With a median of 35 turns, causal chains frequently span multiple chunks. Retrieval should therefore select a conversation before selecting evidence turns.

• **Speaker normalization cannot be assumed.** 661 distinct raw speaker labels were observed. Approximately 94.19% of turns use the canonical Agent/Customer labels, leaving a non-trivial tail of named or special labels.

• **Intent is not an outcome label.** The dataset contains intent and reason-for-call metadata but no universal explicit business-event/outcome field. Treating intent as an outcome would leak a semantic assumption into evaluation.

• **Lexical overlap is ambiguous.** The single-domain dataset contains many semantically related insurance/customer-service interactions. Words such as refund, booking, premium, cancellation, and plan can occur in many unrelated calls, which explains why lexical-only retrieval is vulnerable to contamination.

III. SHARED SYSTEM DESIGN



All systems follow a common contract: immutable raw data, canonical conversation/turn representations, explicit transcript identity, task-aware evidence construction, structured JSON output, grounding checks, and detailed latency tracing. The benchmark is deliberately outside the retrieval implementation so that the same questions and metrics can be applied to every system.

A. Evidence Representation

Each evidence item retains transcript_id, turn identifier, speaker information, original text, normalized text, retrieval score, and retrieval stage. This representation prevents a common RAG failure in which chunks from different conversations are concatenated and then treated as one interaction.

B. Task 1 and Task 2

Task 1—Causal Rationale Extraction targets explicit or strongly supported cause/effect relationships, relevant agent action, mechanism, and evidence turns. **Task 2—Causal Rationale Synthesis** is intended to reconstruct a broader chain consisting of the business event, customer motivation, causal trigger, agent response, outcome, and uncertainty. The latest benchmark archive contains 13 questions and all are labeled Task 1; consequently, no quantitative Task 2 score is reported in the results section.

IV. SYSTEM 1 — DENSE VECTOR RAG

System 1 uses dense semantic retrieval as the baseline: query normalization → embedding → vector search → conversation selection → chunk retrieval → neighboring-turn expansion → evidence gate → LLM generation → grounding validation. The architecture is designed to capture paraphrases and semantic similarity that lexical retrieval may miss.

The most important System 1 lesson was not a model-quality failure but an evidence-assembly failure. Early runs returned several individually similar chunks from different transcript IDs. A causal question then produced a multi-customer answer. The remedy was to retrieve conversations first, then search within the selected conversations, and finally expand around

V. SYSTEM 2 — HYBRID RAG

System 2 combines dense semantic retrieval with lexical retrieval. Candidate sets are merged, conversations are ranked, relevant turns are expanded, and the final evidence pack is sent to the same generation layer. This design targets complementary failure modes: dense retrieval improves semantic recall while lexical retrieval preserves exact business terms, entities, and phrases.

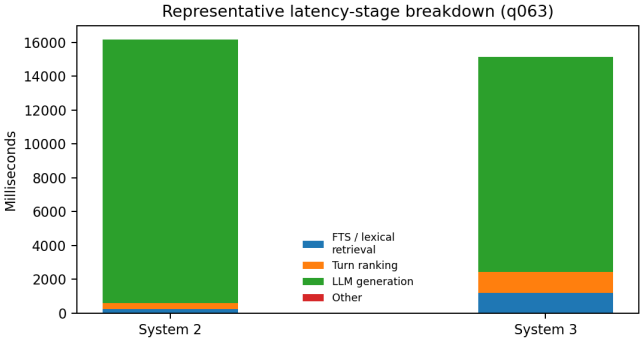


Fig. 7. Representative latency-stage breakdown. For measured System 2/3 calls, generation dominates end-to-end latency; retrieval and ranking are comparatively small.

In the latest 13-query run, System 2 achieved 84.62% conversation recall, 84.62% evidence precision, 62.82% evidence recall, and 15.38% cross-conversation contamination. Mean latency was 15.17 s, with a 12.90 s median and 23.55 s p95. A representative trace shows approximately 15.6 s of generation against roughly 0.6 s for FTS/retrieval and turn ranking combined. The experiment therefore indicates that further retrieval optimization alone cannot remove the dominant latency cost.

The measured generation model was **Qwen3.5:4B-MLX via Ollama** in the latest System 2 run. This keeps the comparison local and reproducible, while making generation a visible bottleneck rather than hiding it inside an external service.

VI. SYSTEM 3 — VECTORLESS RAG

System 3 intentionally removes dense embeddings and vector databases. Its retrieval layer uses lexical/structured signals: FTS/BM25-style matching, phrase and keyword features, metadata, conversation-level aggregation, and local turn expansion. This makes the system interpretable and lightweight, while directly testing whether semantic vectors are necessary for the task.

The latest benchmark reports 53.85% conversation recall, 44.16% evidence precision, 38.46% evidence recall, and 61.54% contamination. Mean latency is 7.90 s. The result demonstrates the expected speed/quality trade-off: lexical retrieval is materially faster than the hybrid system, but repeated vocabulary across unrelated calls causes false matches.

Observed issue	Likely reason	Fix / future action
Global top-k mixed conversations	Chunk-level semantic similarity ignored conversation identity	Conversation-first retrieval and transcript-level grouping
Weak causal evidence at top rank	Closing/filler turns can be semantically similar	Causal query expansion and neighboring-turn expansion
Generation dominated latency	Local 27B generation was much slower than retrieval	Use a smaller MLX model or faster local/cloud generator
Benchmark dependency failure	Missing pydantic_settings in the latest benchmark environment	Pin/install dependency and rerun all 13 queries

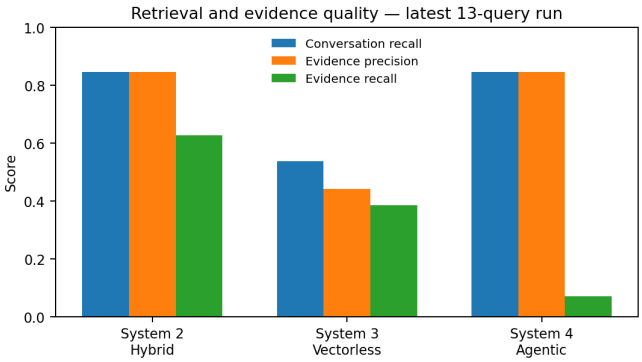


Fig. 8. Latest retrieval/evidence quality. System 2 and System 4 match on conversation recall and evidence precision, while System 4 currently has very low evidence recall.

The most promising fixes were not to add embeddings back, which would violate the experimental purpose, but to strengthen lexical discrimination: field-weighted BM25, exact entity boosts, phrase matching, metadata constraints, speaker-aware scoring, negative-term penalties, and conversation-level score aggregation. These were identified but not fully exhausted within the available time.

VII. SYSTEM 4 — AGENTIC / QUALITY RAG

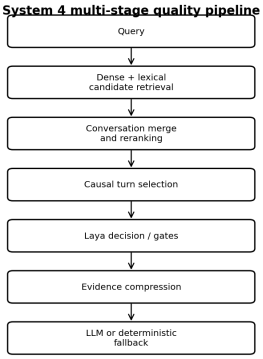


Fig. 9. System 4 uses multi-stage retrieval, feature reranking, causal turn selection, Laya gates, compression, and a final generation/fallback layer.

System 4 broadens candidate recall using dense and lexical retrieval, merges candidate conversations, reranks the candidates using feature-based scoring, selects causally useful turns, applies Laya decision gates, compresses the evidence pack, and validates the final answer. Laya is treated as a deterministic control plane rather than the retriever: it can accept, reject, or gate evidence and answers without silently changing the retrieval experiment.

The latest run is particularly informative. System 4 reaches 84.62% conversation recall, 84.62% evidence precision, and 15.38% contamination—matching System 2 on these three measures—while reporting only 250 ms mean latency. However, evidence recall is just 7.05%. The traces also show **generation_mode = deterministic_fallback**, with approximately zero measured LLM generation time and zero generation tokens per second. Therefore the 250 ms figure is a retrieval/control/fallback latency, not a directly comparable full LLM-powered RAG latency.

This distinction is critical. System 4 has demonstrated an efficient candidate-selection and gating pipeline, but its final quality and end-to-end speed cannot yet be compared fairly against Systems 2 and 3 until a real LLM generation stage is enabled and timed.

A. System 4's Main Failure Mode

The system is over-compressing evidence. It often reduces a conversation to approximately four turns and fewer than 100 context tokens. This explains the combination of high evidence precision and extremely low evidence recall: the selected material is usually relevant, but it does not cover enough of the causal chain.

The appropriate fix is **causal completeness before compression**. For Task 1, the evidence selector should verify that cause, agent action, and effect/outcome are represented before pruning. For Task 2, it should verify business event, motivation, cause, intervention, outcome, and uncertainty. If a required link is absent, the system should expand locally before compression.

VIII. BENCHMARKING METHODOLOGY

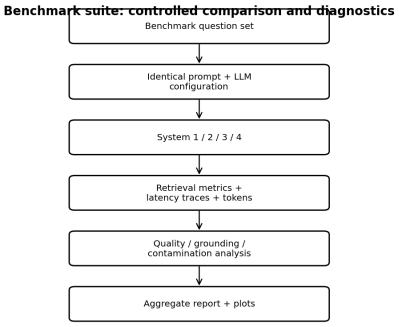


Fig. 10. Benchmark suite used to separate retrieval, answer quality, latency, and diagnostic behavior.

The benchmark records conversation recall, reciprocal rank, evidence precision, evidence recall, cross-conversation contamination, mean/p50/p95 latency, stage-level timings, and per-query traces. The benchmark also retains token counters when available. A key methodological rule is to separate retrieval latency from generation latency; the traces show that generation can be orders of magnitude larger than retrieval.

TABLE III. CORE BENCHMARK METRICS

Metric	Purpose
Conversation Recall	Whether the correct transcript is retrieved.
MRR	How highly the correct conversation is ranked.
Evidence Precision	Fraction of selected evidence judged relevant.
Evidence Recall	Fraction of reference evidence recovered.
Contamination	Whether unrelated transcripts enter the evidence set.
p50 / p95 latency	Typical and tail response time.
TTFT / generation	Model responsiveness and generation cost.
Grounding status	Whether generated claims are supported.

IX. LATEST BENCHMARK RESULTS

TABLE IV. LATEST 13-QUERY FULL RUN (RUN 20261002T103441Z_2306)

System	Fail.	Conv. R@1	MRR	Ev. Prec.	Ev. Rec.	Contam.	Mean
S1 Dense	13/13	—	—	—	—	—	—
S2 Hybrid	0/13	84.62%	0.846	84.62%	62.82%	15.38%	15.17 s
S3 Vectorless	0/13	53.85%	0.464	44.16%	38.46%	61.54%	7.90 s
S4 Agentic	0/13	84.62%	0.846	84.62%	7.05%	15.38%	250 ms

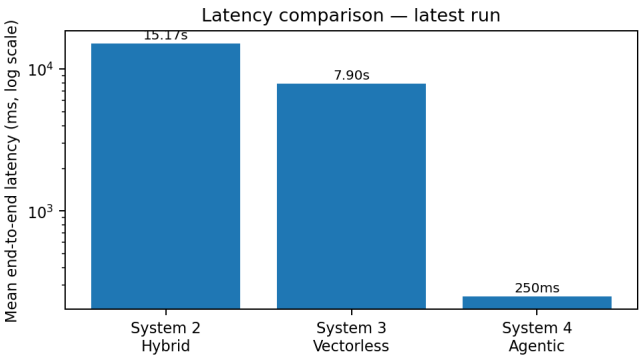


Fig. 11. Mean latency on the latest run. System 4 is not directly comparable because its LLM stage is bypassed by a deterministic fallback.

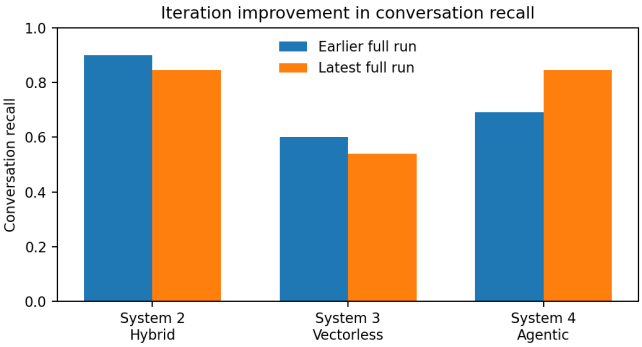


Fig. 12. Iteration-to-iteration change in conversation recall from an earlier full run to the latest full run.

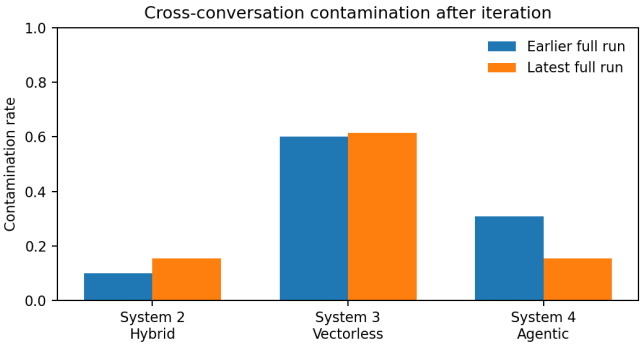


Fig. 13. Cross-conversation contamination improved substantially for System 4, while System 3 remained vulnerable to lexical ambiguity.

A. What Improved

• **System 2:** failures fell from 3/13 to 0/13 and mean latency decreased by about 12.4%, although conversation recall and precision moved from 90.0% to 84.62%.

• **System 3:** failures fell from 3/13 to 0/13 and mean latency decreased by about 8.8%, while contamination remained high at 61.54%.

• **System 4:** conversation recall increased from 69.23% to 84.62% and contamination fell from 30.77% to 15.38%. The improvement came with a modest increase in control-pipeline latency and did not solve evidence recall.

X. ERROR ANALYSIS AND ROOT CAUSES

TABLE V. FAILURE MODES, CAUSES, AND FIXES

Failure pattern	Observed evidence	Likely cause	Priority
Cross-conversation synthesis	Strong in early dense and vectorless runs	Global chunk ranking ignores transcript identity; lexical overlap across insurance calls	High
Insufficient causal context	System 4 evidence recall 7.05%	Aggressive turn selection/compression	High
LLM latency	S2 mean 15.17 s; S3 mean 7.90 s	Local generation dominates request time	High
Vectorless ambiguity	S3 contamination 61.54%	Shared terms and weak entity/metadata discrimination	High
Invalid causal abstractions	Development examples copied utterances as cause/effect	Grounding validator checks source support, not causal semantics	High
Task routing instability	Interactive Task 2 tests returned Task 1 labels	Task classifier/output contract not sufficiently isolated	High
System 1 unavailable	13/13 benchmark failures	Environment dependency missing	Immediate
System 4 timing anomaly	LLM generation ~0 ms	Deterministic fallback bypassed actual generator	Immediate

A. Grounding Is Not Causal Correctness

A particularly important lesson is that a response can be fully grounded in the literal sense while still being causally wrong. For example, a system can quote a customer complaint as the 'cause' and a later agent utterance as the 'effect' even when the conversation does not establish that causal relationship. The final validator should therefore test both *source grounding* and *causal validity*. Deterministic checks should reject identical cause/effect strings, non-chronological evidence, question-only effects, and claims whose relation is not supported by the selected turns.

XI. EXPERIMENTS REMAINING UNDER TIME CONSTRAINT

• **Enable measured LLM generation in System 4.** Configure the same Qwen3.5 MLX model used by Systems 2/3 and measure TTFT, output tokens, tokens/s, and total latency.

• **Run a larger benchmark.** The current 13-query set is diagnostic. A final study should use at least 100 queries for tuning and preferably 500–1,000+ for the final comparison, with balanced Task 1/Task 2 coverage.

• **Complete System 1.** Fix the missing pydantic_settings dependency and rerun the same benchmark without changing questions or evaluation code.

• **Improve System 4 evidence completeness.** Add a causal-chain checklist before compression and adaptive ±N turn expansion when a required causal slot is missing.

• **Strengthen System 3 without vectors.** Tune BM25 field weights, exact phrase/entity matching, speaker-aware features, metadata constraints, negative terms, and conversation-level aggregation.

• **Model-size ablation.** Compare Qwen3.5 4B MLX and 9B MLX under identical prompts and contexts to separate retrieval effects from generator capacity.

• **Answer-quality evaluation.** Add human or calibrated LLM-judge labels for causal correctness, completeness, attribution, uncertainty, and unsupported claims. Current benchmark metrics primarily evaluate retrieval/evidence behavior.

• **Token efficiency.** Record input/output tokens for every provider and report quality per 1k context tokens and quality per second once all generators expose consistent counters.

• **Task 2 benchmark coverage.** Build a dedicated Task 2 question set rather than relying on Task 1 questions with a different prompt.

XII. PRODUCTION ARCHITECTURE AND DEPLOYMENT

The project was structured around modular retrievers, evidence construction, LLM adapters, validation, benchmarking, and a Streamlit-facing application layer. This separation allows the same LLM interface to be used with local Ollama models, the earlier local Bonsai endpoint, or a permitted cloud provider without rewriting retrieval code.

TABLE VI. RECOMMENDED PRODUCTION LAYERING

Layer	Responsibility
Data	Immutable source, normalization, sampling, metadata
Retrieval	Dense, hybrid, or lexical candidate generation
Reranking	Conversation and turn relevance refinement
Evidence	Expansion, causal completeness, compression
Decision	Laya gates and deterministic safety checks
Generation	Bonsai / Ollama / other adapter
Validation	Grounding, chronology, transcript identity, causal consistency
Benchmark	Metrics, traces, latency, failures, reports
UI	Streamlit query interface and diagnostics

Laya is most useful as a fast decision/control layer: task routing, evidence sufficiency, answer gating, and policy decisions. It should not silently replace the retrieval method

being evaluated. This preserves experimental validity while still allowing production-time control decisions.

XIII. CONCLUSION

The project demonstrates that causal conversational RAG is primarily an evidence-management problem. Dense retrieval alone is not sufficient if conversation identity is lost; lexical retrieval alone is fast but vulnerable to shared business vocabulary; and multi-stage agentic retrieval can improve candidate precision dramatically, but aggressive compression can destroy the causal chain.

The latest results provide three concrete engineering directions. First, System 2 provides a strong evidence-quality baseline but needs faster generation. Second, System 3 provides a meaningful vectorless baseline and exposes the cost of lexical ambiguity. Third, System 4 has the strongest architectural potential for a low-latency quality pipeline, but it must replace deterministic fallback with a real measured generator and substantially increase evidence recall before its latency and answer quality can be compared fairly.

The resulting framework is therefore not just a collection of four RAG implementations. It is a controlled experimental platform in which retrieval strategy, evidence construction, decision control, generation, and validation can be independently measured and improved.

REFERENCES

[1] Internal hackathon problem statement, “Causal Rationale Extraction and Synthesis from Conversational Data on Business Events,” dataset and task specification.

[2] Internal conversational transcript dataset, 19,621 records, materialized locally as final_transcripts(2).json.

[3] Project EDA notebooks: task1_dataset_specific_EDA.ipynb and task1_dataset_specific_traditional_rag.ipynb.

[4] Benchmark Suite artifacts, latest full run 20261002T103441Z_2306: aggregate_metrics.json, quality_metrics.json, latency_metrics.json, contamination_metrics.json, traces.jsonl, results.jsonl.

[5] Project architecture specifications: AGENTS_SYSTEM_2.md, AGENTS_SYSTEM_3.md, and the System 4 agentic/quality design specification.